A total-variation floor and a rank-three classification for the parity-compressed sector of a log-singular Toeplitz form with a prime-power comb

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Abstract

This paper is about the near-degeneracy behind a Toeplitz spectral floor: a total-variation bound, three linear functionals, and a priced table of what the standard unconditional toolbox reaches. The object is the reflection-odd compression A of a finite Toeplitz form whose lag sequence is a logarithmic profile minus an explicit comb, and the question is how small the Rayleigh quotient of A can be on the low end of the parity spectrum. Four results are proved. First, a *total-variation floor*: every trial vector whose leading parity coordinate is normalised to one has lag weights of total variation at least $1/\mu_1$, where $\mu_1 = 4\sin^2(\pi/(2h+1))$ is the spectral gap of the parity compression of the second difference; the floor is closed, involves no comb node, no taper and no truncation, and it prices the entire trial-vector route at $O(h^{-2})$ for every vector of bounded value. Second, a *gauge rigidity* statement: the leading-coordinate normalisation pins only the scale of the trial direction, both the value and the total variation are homogeneous of degree two, hence their ratio is a function of the ray and takes the same range in every reparametrisation of the entry spectrum — no sector change removes the floor, and a shifted sector conserves it exactly, floor times transfer being $1/\mu_1$ identically. Third, an exact *rank-three classification*: the two-by-two raw Gram block of the low parity modes splits as an archimedean block minus a comb block, the determinant polarises into three terms, and the comb–comb term is the polarisation of the determinant on symmetric two-by-two matrices — a form of rank three and signature $(1, 2)$ — so the comb enters the determinant only through *three* linear functionals of its weights, for every window and every truncation. Fourth, a classification of what this costs: any majorant of the comb–comb term built from a norm of the weight vector must dominate the supremum of a rank-three quadratic over the image of that norm ball, which is where the priced routes lose. The remaining question — whether the two rows of the block become collinear at a rate uniform in the window index — is stated as an open problem about a finite bilinear form, with the measured evidence, the placement discriminators and the priced routes reported as measurements on declared surfaces. The arithmetic enters the theorems only as the source of the comb positions and weights: every statement proved here holds for an arbitrary finite comb.

1 Introduction

1.1 The problem, and why the direction is hard

Let $c = (c_m)_{m=0}^{M-1}$ be a real lag sequence, let $T_M(c)$ be its symmetric Toeplitz matrix, and let A be the compression of $T_M(c)$ to the reflection-odd sector of the window (Section 2). The question this paper is about is a lower bound for the Rayleigh quotient of A on the low end of a fixed orthonormal mode family — equivalently, an upper bound for the extent to which the quadratic form $v \mapsto v^{\mathsf{T}}Av$ can cancel along a smooth direction.

The classical route from a Toeplitz form to a spectral bound is the symbol. If $\ell \leq \sigma$ pointwise then $T_M(\ell) \preceq T_M(\sigma)$ by Parseval, and a positive pointwise minorant of the symbol transfers to a positive floor for the form [20, 9, 13, 6, 7]. For the class considered here that route is unavailable, and not marginally so: the symbol is a logarithmic profile minus an explicit cosine comb, the comb dips are numerous and narrow, and on every window of the declared surface of Section 8 the infimum of the symbol is negative, while the form restricted to the parity sector is positive. The smallest Rayleigh quotient is therefore not produced by a small value of the symbol at some frequency; it is produced by *cancellation inside the quadratic form*, along a direction that lives in the low modes. A pointwise minorant is by construction blind to which vector realises the minimum.

The natural substitute is a trial-vector route: exhibit an explicit direction, evaluate the form on it, and price the evaluation. This paper is a classification of that route. Its first two results say that the route has a closed price which no reparametrisation removes; its second two say that the object which remains after the price is paid is a quadratic polynomial in three linear functionals of the comb weights, and that this is exactly why the unconditional toolbox — size budgets, symbol arguments, sectors and gauges, splits, perturbation theory, the bilinear sieve box — does not reach it. What is *not* claimed anywhere is a bound on the arithmetic: the three functionals are finite sums over the comb, all four results hold for an arbitrary finite comb, and the one remaining question is stated as an open problem in Section 7.

The companion paper [10] treats the same window form from the opposite direction — a certified lower bound for a Galerkin defect on a nested ladder — and we cite it for the window form and the comb class rather than repeating them.

1.2 Contribution

- (C1) **The total-variation floor, and the price it fixes** (Theorem 3.1 with Corollaries 3.2–3.4). For every trial vector whose leading parity coordinate is normalised to one, the lag weights w of the associated window vector satisfy $\mathrm{TV}(w) \geq |w_0| = \|v\|^2 \geq 1/\mu_1 \geq N^2/(4\pi^2)$, where $\mu_1 = 4\sin^2(\pi/N)$, $N = 2h+1$, is the spectral gap of the parity compression L_P of the second difference [13]. The proof is four steps and uses no property of the comb whatsoever. The load-bearing consequence is a *price*: the ratio of the value to the total variation is at most μ_1 times the value, so every trial vector of bounded value certifies a ratio $O(h^{-2})$, and the boundary-free summation by parts [1] can certify nothing better.
- (C2) **Gauge rigidity of the normalisation** (Theorem 4.1 with Propositions 4.3–4.4 and Corollary 4.2). The normalisation pins only the scale of the trial direction; value and total variation are homogeneous of degree two; hence their ratio is a function of the ray, and its range over admissible vectors is the *same* for every entry spectrum. A shifted spectrum does flatten the floor, but the transfer factor that returns the bound to the original scale conserves the product exactly: floor times transfer is $1/\mu_1$ at every shift. So (C1) cannot be reparametrised away, and the parity compression is not a convenience — Proposition 4.4 shows it is what makes the normalisation well posed at all.
- (C3) **The exact bilinear form and the rank-three classification** (Section 6: Propositions 6.1 and 6.2, Theorem 6.3, Corollary 6.4). The two-by-two raw Gram block of the two lowest parity modes splits exactly as $\widehat{A}_2 = B - S$ with B archimedean and S a weighted sum of comb reads; the determinant polarises as $\det \widehat{A}_2 = \det B - D(B, S) + \det S$; and D is the polarisation of det on symmetric two-by-two matrices, a form of rank three and signature $(1, 2)$ in the coordinates (X_{11}, X_{22}, X_{12}). Consequently the comb–comb term is $S_{11}S_{22} - S_{12}^2$: a quadratic polynomial in *three* linear functionals of the comb weights, for every window, every h and every truncation. Corollary 6.4 turns this into the statement about majorants that the priced routes of Section 7 run into.
- (C4) **A priced table and one open problem** (Section 7). What remains after (C1)–(C3)

is a single question about a finite bilinear form: is the near-collinearity of the two rows of $\widehat{A}_2$ bounded uniformly in the window index? It is stated as Problem 7.1 in three equivalent forms, with the priced routes, the measured rates, the density curve and the placement discriminators reported as measurements on declared surfaces — never as an approach to, a weakening of, or a consequence of any conjecture.

Two of these four are theorems; two are classifications supported by theorems together with declared measurements. Which is which is recorded in Section 1.3 rather than left to the reader.

1.3 Typing of the results

Every numbered statement of this paper carries exactly one of three types, and the type is part of the statement.

proved
: A written proof is given (or, for classical facts, cited). The statement holds for every window and every finite comb within its stated hypotheses. Where a hypothesis is itself of a different type — positive definiteness of A on the parity sector, for instance — the statement is *conditional* and says so.

certified per window
: The statement is verified on each window of a declared finite surface by a computation whose failure mode is loud (a completed Cholesky factorisation, an exact identity residual against a declared floor). No quantifier over windows is claimed.

measured
: The statement is a number, a fitted exponent, or a controlled experiment on a declared finite surface, reported as such. Fitted exponents are never used as bounds.

Table 1: Typing of every numbered statement. "Conditional" marks a proved implication whose hypothesis is of a weaker type; the hypothesis is then named in the statement.

statement	type	remark
Section 2 — setting and scaffolding		
L1 interlacing (Lem. 2.3)	proved (cited)	Cauchy [3], Courant–Fischer [5]
L2 Schur nesting (Lem. 2.4)	proved	Schur [19], Haynsworth [11]
L3 constrained minimum (Lem. 2.5)	proved	requires $G_k \succ 0$
gauge law (Lem. 2.9)	proved	$\sum_d w_d = 0$, unconditional
Abel swap (Lem. 2.10)	proved	boundary-free, $w_M = 0$
C5 cascade forms (Prop. 2.6)	proved	prefix / minor / regression
C6 diagonal invariance (Lem. 2.7)	proved	one line
B4 cascade direction (Prop. 2.8)	*conditional*	proved given $\widehat{A} \succ 0$; the hypothesis is certified per window
Section 3 — the total-variation floor		
A the floor (Thm. 3.1)	proved	comb-free, taper-free
A1 the price (Cor. 3.2)	proved	the load-bearing corollary

continued on the next page

statement	type	remark
A2 Abel/Hölder (Cor. 3.3)	proved	
A-bis (Cor. 3.4)	proved	no normalisation needed
A4 negative control (Rem. 3.5)	proved	one line
A3 sharpness (Meas. 3.6)	measured	μ_1 TV $\in [5.00, 23.20]$ on Surface II
Section 4 — gauge rigidity		
B homogeneity (Thm. 4.1)	proved	
B1 floor $\times$ transfer (Prop. 4.3)	proved	pure shift
B2 the null mode (Prop. 4.4)	proved	closed spectra
no escape (Cor. 4.2)	proved	
sector battery (Meas. 4.5)	certified per window	$35 \times 6 \times 27$, residual $\leq 8.8 \cdot 10^{-10}$
Section 5 — the two-dimensional reduction		
C1 exact $K = 2$ (Prop. 5.1)	proved	
C2 pivot identity (Prop. 5.2)	proved	
C3 union identity (Prop. 5.3)	proved	
C7 Wronskian (Lem. 5.5)	proved	the corner entry 3 closes it
C8 maximal minors (Lem. 5.6)	proved	Lagrange [15]
C9 determinant reading (Lem. 5.7)	proved	uses no positivity
C10 Gershgorin (Lem. 5.8)	proved	uniform in the window
C11 det-collapse (Prop. 5.9)	proved / *conditional*	the closed form is unconditional; the harmonic-mean bound needs $\widehat{A}_2 \succ 0$
Section 6 — bilinear form and rank		
D0 exact split (Prop. 6.1)	proved	needs an admissible resolution
D1 polarisation (Prop. 6.2)	proved	
D Theorem D, rank three (Thm. 6.3)	proved	every h, every truncation
D3 majorants (Cor. 6.4)	proved	declared class of majorants
D4 Type II rank (Meas. 6.5)	measured	bandwidth $\Rightarrow$ η-rank lemma: open writing task
D5 reference window (Meas. 6.6)	measured	the three-term cancellation
Section 7 — the open problem		
Problem 7.1	open	three equivalent forms
rates (Meas. 7.2)	measured	fitted exponents, declared surfaces
priced routes (Meas. 7.3)	proved per route / measured exponent	each bound unconditional, each exponent a fit
density curve (Meas. 7.4)	measured	undecidability is part of the claim
placement (Meas. 7.5)	measured	controlled experiment

Table 1 is the first thing to read. Three entries deserve emphasis. Proposition 2.8 and the harmonic-mean half of Proposition 5.9 are *conditional*: they are proved implications whose

symbol	meaning
$D,\ M \in 2\mathbb{N},\ \alpha = MD/2,\ h = M/2,\ N = 2h+1$	cell width, window dimension, half-width, sector dimension, parity period
$T_M(c) = (c_{\lvert r-r'\rvert})_{r,r'=0}^{M-1}$	symmetric Toeplitz matrix of the lag sequence c
$\Pi \in \mathbb{R}^{M\times h}$, $\Pi e_r = \frac{1}{\sqrt{2}}(e_r - e_{M-1-r})$	reflection-odd isometry, $\Pi^\mathsf{T}\Pi = I_h$
$A = \Pi^\mathsf{T} T_M(c)\Pi$, $A_{rr'} = c_{\lvert r-r'\rvert} - c_{M-1-r-r'}$	the odd window form
$a(\cdot)$; $S_D(s,u)$; ϱ_j, u_j	archimedean kernel; symmetrised hat; comb weights and node positions
L_P	odd compression of the lag sequence $(2,-1,0,\dots)$: tridiagonal $2I - E - E^\mathsf{T}$ with corner entry 3
$\mu_k = 4\sin^2(k\pi/N)$, $(t_k)_r = \frac{2}{\sqrt{N}} \sin\frac{2\pi k(r+1)}{N}$	parity eigenvalues and orthonormal parity modes, $k = 1,\dots,h$
$\widehat{a}_{ij} = t_i^\mathsf{T} A t_j$, $\widehat{A}_K = (\widehat{a}_{ij})_{i,j\le K}$	the raw Gram block
$\mathcal{M} = \operatorname{diag}(\mu_1,\dots,\mu_K)$, $G = \mathcal{M}^{-1/2}\widehat{A}_K\mathcal{M}^{-1/2}$	the normalised K-block
G_k, $g_k = e_1^\mathsf{T} G_k^{-1} e_1$, $s = \mu_1 t_1^\mathsf{T} A^{-1} t_1$	leading blocks, the ladder, the entry functional
x admissible: $x \in \mathbb{R}^K$, $x_1 = 1$	trial vector
$a_k = x_k/\sqrt{\mu_k}$, $v = \sum_{k\le K} a_k t_k$	coefficient vector and window vector
$Q(x) = x^\mathsf{T} G x = v^\mathsf{T} A v$	the value
$w_0 = \mathrm{Ac}_0(v)$, $w_d = 2\,\mathrm{Ac}_d(v) - \mathrm{Cv}_{M-1-d}(v)$ $(d \ge 1)$, $w_M := 0$	lag weights; Ac autocorrelation, Cv self-convolution
$(\Delta w)_d = w_d - w_{d+1}$, $\mathrm{TV}(x) = \lVert \Delta w \rVert_1$	lag increments and total variation
$C_d = \sum_{e\le d} c_e$, $C_{-1} := 0$	partial sums of the lag sequence
B, S, $r_{12}^2 = \widehat{a}_{12}^2/(\widehat{a}_{11}\widehat{a}_{22})$, $\nu_1 \ge \nu_2$	archimedean block, comb block, squared correlation, eigenvalues of $\widehat{A}_2$
$D(P,Q) = P_{11}Q_{22} + P_{22}Q_{11} - 2P_{12}Q_{12}$	polarisation of det on symmetric 2×2 matrices

Table 2: Notation, fixed for the whole paper.

hypothesis is positive definiteness of the raw Gram block, which on the surfaces of Section 8 is a numerical fact about finitely many finite matrices and nothing more. Measurement 6.5 is the one place where a statement of the source material does not convert: "effective rank $O(1)$" is not a theorem, the substitute (a bandwidth $\Rightarrow$ η-rank lemma with a declared truncation error) is not written in this revision, and the row therefore stays measured.

1.4 Notation

Notation is fixed once, in Table 2, and not varied. Three choices need a word. The symbol Π denotes the reflection-odd isometry, so that B can be reserved for the archimedean Gram block of Section 6; $\mathcal{M}$ (calligraphic) denotes $\operatorname{diag}(\mu_1,\dots,\mu_K)$, so that M (upright) keeps its meaning as the window dimension; and the comb weights, written μ_j in [10], are written ϱ_j here, because μ_k is needed for the parity eigenvalues. The archimedean kernel is written $a(\cdot)$ and always carries its argument; the coefficient vector of a trial vector is written a and never does.

$X \succeq 0$ and $X \succ 0$ denote positive semidefiniteness and definiteness, $\preceq$ the Löwner order, $\lVert\cdot\rVert$ the Euclidean norm, e_k the k-th coordinate vector, $|X|$ the entrywise absolute value of a matrix, and sgn the entrywise sign. For $v \in \mathbb{R}^h$ the autocorrelation and self-convolution are

$$\mathrm{Ac}_d(v) = \sum_r v_r v_{r+d} \quad (d \ge 0), \qquad \mathrm{Cv}_j(v) = \sum_{r+r'=j} v_r v_{r'}, \tag{1}$$

both sums over indices in $\{0,\dots,h-1\}$ and empty sums being zero. *Admissible* always means $x_1 = 1$.

2 Setting

2.1 The odd window form and the comb class

We use the window form and the comb class of [10] verbatim and recall only what is needed. Fix a cell width $D > 0$ and an even M, put $\alpha = MD/2$, $h = M/2$, and let Π be the reflection-odd isometry of Table 2. Given a real lag sequence $c = (c_m)_{m=0}^{M-1}$ the odd window form is

$$A = \Pi^{\mathsf{T}} T_M(c)\,\Pi \in \mathbb{R}^{h\times h}, \qquad A_{rr'} = c_{|r-r'|} - c_{M-1-r-r'}. \tag{2}$$

The class we verify on is the comb class of [10, Def. 2.6]: with $a(\cdot)$ the archimedean kernel of that definition, a finite node set $\{(u_j,\varrho_j)\}_j$ with $u_j > 0$, $\varrho_j > 0$, and $S_D(s,u) = \frac{1}{2}[\Delta_D(s-u) + \Delta_D(s+u)]$ the symmetrised hat built from the unit-height triangle $\Delta_D(t) = \max\{0, 1-|t|/D\}$ of half-width D,

$$c_m = a(mD) - \sum_{j:\,u_j\le 2\alpha} \varrho_j\, S_D(mD,u_j), \qquad m = 0,\dots,M-1. \tag{3}$$

The *prime-power instance* is $u_j = \log n_j$ over the prime powers $n_j = p^\nu$ in increasing order with $\varrho_j = 2\Lambda(n_j)/\sqrt{n_j}$, Λ the von Mangoldt weight. By linearity of $c \mapsto A$ the split of (3) induces a split

$$A = A_{\text{arch}} - A_{\text{comb}}, \qquad A_{\text{comb}} = \sum_j \varrho_j\, A^{(j)}, \quad A^{(j)} = \Pi^{\mathsf{T}} T_M(S_D(\cdot D, u_j))\Pi. \tag{4}$$

Throughout, a *resolution* D is called *admissible* if $D < \min_j u_j$; for the prime-power instance this is implied by the admissibility convention of [10, Def. 5.1], which fixes D as a fraction of the smallest log-gap in the run.

Remark 2.1 (the arithmetic is not used)**.** Theorems 3.1, 4.1 and 6.3, and every lemma, proposition and corollary of Sections 2–6, hold for an *arbitrary* finite comb: the node positions u_j and the weights ϱ_j enter only as the data of (3), and no property of the sequence (u_j) or (ϱ_j) is used. Where the prime-power instance is mentioned it is as the instance the measurements of Sections 7–8 were taken on, and the only arithmetic facts consumed anywhere are the two classical gap inequalities used in [10] to check that a resolution is admissible [4, 18]. No arithmetic statement is claimed.

2.2 The parity spectrum

Let $c^L = (2,-1,0,\dots,0)$ and let L_P be the odd window form (2) of c^L.

Lemma 2.2 (the parity compression and its spectrum)**.** *L_P is the tridiagonal matrix $2I - E - E^{\mathsf{T}}$ on $\mathbb{R}^h$ with corner entry $(L_P)_{h-1,h-1} = 3$; equivalently, L_P is the second-difference form with a Dirichlet condition at the outer end and the odd reflection $u_h = -u_{h-1}$ at the inner end. Its eigenpairs are*

$$\mu_k = 4\sin^2\Big(\frac{k\pi}{N}\Big), \qquad (t_k)_r = \frac{2}{\sqrt{N}}\sin\frac{2\pi k(r+1)}{N}, \qquad k = 1,\dots,h,\ r = 0,\dots,h-1, \tag{5}$$

with $N = 2h+1$, and $\{t_k\}_{k=1}^h$ is an orthonormal basis of $\mathbb{R}^h$. In particular $\mu_1 = 4\sin^2(\pi/N)$ is the smallest eigenvalue and

$$\frac{1}{\mu_1} \;\ge\; \frac{N^2}{4\pi^2} \;>\; \frac{h^2}{\pi^2}. \tag{6}$$

Proof. For c^L the Hankel part of (2) is $-c^L_{M-1-r-r'}$, which is nonzero only if $M-1-r-r' \in \{0,1\}$, i.e. only if $r+r' \in \{2h-2, 2h-1\}$; since $r, r' \le h-1$ the only possibility is $r = r' = h-1$, contributing $-c^L_1 = +1$. Hence L_P is tridiagonal with diagonal 2 except for the corner entry $2+1=3$, which is the first assertion. Reading the rows as a recurrence, row 0 is $2u_0 - u_1 = \mu u_0$, i.e. the convention $u_{-1} = 0$, and row $h-1$ is $-u_{h-2} + 3u_{h-1} = \mu u_{h-1}$, i.e. $-u_{h-2} + 2u_{h-1} - u_h = \mu u_{h-1}$ with $u_h = -u_{h-1}$.

For the eigenpairs, put $u_r = \sin(\omega(r+1))$. Then $u_{-1} = 0$ and the interior recurrence holds with $\mu = 2 - 2\cos\omega = 4\sin^2(\omega/2)$. The inner condition $u_h = -u_{h-1}$ reads $\sin(\omega(h+1)) + \sin(\omega h) = 2\sin(\frac{\omega N}{2})\cos(\frac{\omega}{2}) = 0$, and since $0 < \omega < \pi$ the second factor is positive, so $\omega N/2 = k\pi$, i.e. $\omega = 2k\pi/N$ and $\mu = 4\sin^2(k\pi/N)$. The values $k = 1, \ldots, h$ give h distinct eigenvalues, hence all of them. For the normalisation, $\sum_{r=0}^{N-1} \sin^2(2\pi kr/N) = N/2$, the term $r = 0$ vanishes, and the remaining $2h$ terms pair up as $r \leftrightarrow N-r$ with equal values, so $\sum_{r=1}^{h} \sin^2(2\pi kr/N) = N/4$ and the factor $2/\sqrt{N}$ normalises. Orthogonality is symmetry of L_P together with simplicity of the spectrum. Finally $\sin t \le t$ on $[0, \pi/2]$ gives $\mu_1 \le 4\pi^2/N^2$, which is (6); the last inequality is $N = 2h+1 > 2h$. □

The spectrum (5) is the classical Kac–Murdock–Szegő spectrum of the second difference in the reflection-odd sector [13], and μ_1 is its spectral gap. Everything in Sections 3 and 4 is driven by (6) and by nothing else.

2.3 The raw Gram block, the normalised block, and the ladder

Fix $K \le h$ (the verified surfaces use $K = 16$; nothing below depends on the value). With $\{t_k\}$ of Lemma 2.2 put

$$\widehat{a}_{ij} = t_i^\mathsf{T} A t_j, \qquad \widehat{A}_K = (\widehat{a}_{ij})_{i,j \le K}, \qquad G = \mathcal{M}^{-1/2} \widehat{A}_K \mathcal{M}^{-1/2}, \qquad \mathcal{M} = \operatorname{diag}(\mu_1, \ldots, \mu_K), \tag{7}$$

write G_k for the leading $k \times k$ block of G, and

$$g_k = e_1^\mathsf{T} G_k^{-1} e_1 \quad (k = 1, \ldots, K), \qquad s = \mu_1\, t_1^\mathsf{T} A^{-1} t_1. \tag{8}$$

The following three facts are classical; we record them in the form used later.

Lemma 2.3 (interlacing)**.** *Let $X \in \mathbb{R}^{n\times n}$ be symmetric and $P \in \mathbb{R}^{n\times m}$ an isometry. Then the eigenvalues of $P^\mathsf{T} X P$ interlace those of X; in particular $\lambda_{\min}(X) \le \lambda_{\min}(P^\mathsf{T} X P)$. Consequently a K-truncated Rayleigh minimum is an upper bound for $\lambda_{\min}$ and never a lower bound.*

Proof. Cauchy's interlacing theorem [3] in the Courant–Fischer form [5]. □

Lemma 2.4 (Schur nesting)**.** *Let $X \succ 0$ be partitioned as $\left[\begin{smallmatrix} \xi & b^\mathsf{T} \\ b & Y \end{smallmatrix}\right]$ with $\xi \in \mathbb{R}$. Then $Y \succ 0$ and*

$$\frac{1}{(X^{-1})_{11}} = \xi - b^\mathsf{T} Y^{-1} b, \qquad (X^{-1})_{11} = \frac{\det Y}{\det X}. \tag{9}$$

Proof. The first identity is the Schur complement of the $(1,1)$ pivot [19, 11]; the second is Jacobi's formula for a cofactor of the inverse, i.e. Cramer's rule. □

Lemma 2.5 (constrained minimum)**.** *Let $X \succ 0$ be $k \times k$. Then*

$$\min\{u^\mathsf{T} X u : \ u \in \mathbb{R}^k, \ u_1 = 1\} = \frac{1}{e_1^\mathsf{T} X^{-1} e_1}, \tag{10}$$

attained exactly at $u^\star = X^{-1}e_1/(e_1^\mathsf{T} X^{-1} e_1)$. Equivalently, every *admissible u gives $u^\mathsf{T} X u \ge 1/(e_1^\mathsf{T} X^{-1} e_1)$: an explicit trial vector is a certified lower bound for the reciprocal, never an upper one.*

Proof. Write $u = u^\star + \delta$ with $\delta_1 = 0$. Since $Xu^\star = e_1/(e_1^\top X^{-1} e_1)$, the cross term is $2\delta^\top X u^\star = 2\delta_1/(e_1^\top X^{-1} e_1) = 0$, so $u^\top X u = (u^\star)^\top X u^\star + \delta^\top X \delta$ and $(u^\star)^\top X u^\star = (u^\star)^\top e_1/(e_1^\top X^{-1} e_1) = 1/(e_1^\top X^{-1} e_1)$ because $u_1^\star = 1$. As $X \succ 0$ the remainder is positive unless $\delta = 0$. □

Proposition 2.6 (three forms of the ladder)**.** *Assume $G \succ 0$ and let $G = LL^\top$ be its Cholesky factorisation, $y = L^{-1} e_1$. Then for every $k \le K$:*

$$\text{(prefix)} \qquad g_k = \sum_{j \le k} y_j^2; \tag{11}$$

$$\text{(minor)} \qquad \frac{1}{g_k} = \frac{\det \widehat{A}_{[1..k]}}{\mu_1 \det \widehat{A}_{[2..k]}} \qquad (\det \widehat{A}_{[2..1]} := 1); \tag{12}$$

$$\text{(regression)} \qquad \frac{g_k}{g_1} = \frac{1}{1 - R_k^2}, \qquad R_k^2 = \frac{b^\top G_{HH}^{-1} b}{G_{11}}, \tag{13}$$

where in (13) *the block G_k is partitioned as $\left[\begin{smallmatrix} G_{11} & b^\top \\ b & G_{HH} \end{smallmatrix}\right]$ with $H = \{2, \dots, k\}$. In particular $k \mapsto g_k$ is nondecreasing, strictly increasing whenever $y_k \ne 0$, and $R_k^2 \in [0, 1)$ is the squared multiple correlation of the first coordinate on the remaining $k - 1$.*

Proof. Prefix. For a Cholesky factorisation the leading $k \times k$ block of G is $L_k L_k^\top$ with L_k the leading block of L, so $g_k = e_1^\top L_k^{-\top} L_k^{-1} e_1 = \|L_k^{-1} e_1\|^2$; and since L is lower triangular, forward substitution gives $L_k^{-1} e_1 = (L^{-1} e_1)_{1:k} = y_{1:k}$. Monotonicity is immediate.

Minor. By (9), $g_k = \det G_{[2..k]} / \det G_{[1..k]}$. Substituting $G = \mathcal{M}^{-1/2} \widehat{A}_K \mathcal{M}^{-1/2}$ gives $\det G_{[1..k]} = \det \widehat{A}_{[1..k]} / \prod_{j \le k} \mu_j$ and $\det G_{[2..k]} = \det \widehat{A}_{[2..k]} / \prod_{2 \le j \le k} \mu_j$, and all factors cancel except μ_1, which yields (12).

Regression. By (9), $1/g_k = G_{11} - b^\top G_{HH}^{-1} b = G_{11}(1 - R_k^2)$, while $g_1 = 1/G_{11}$; divide. □

Lemma 2.7 (diagonal invariance of the ladder gain)**.** *For every positive diagonal $\Lambda = \operatorname{diag}(d_1, \dots, d_K)$ the gain $g_k/g_1 = G_{11}\,(G_k^{-1})_{11}$ is unchanged under $G \mapsto \Lambda G \Lambda$.*

Proof. $(\Lambda G \Lambda)_{11} = d_1^2 G_{11}$ and $\big((\Lambda G \Lambda)_k^{-1}\big)_{11} = d_1^{-2} (G_k^{-1})_{11}$; the product is invariant. □

Lemma 2.7 localises the phenomenon: the gain does not see the $\mu^{-1/2}$ normalisation of (7) at all, so whatever makes g_K/g_1 large is a property of the raw Gram block $\widehat{A}_K$ alone, and the parity spectrum enters it only through the choice of the modes.

Proposition 2.8 (direction of the ladder)**.** *Assume $\widehat{A}_h \succ 0$, i.e. $A \succ 0$ on the parity sector. Then*

$$s \;\ge\; g_K \;\ge\; g_1 = \frac{1}{G_{11}}, \qquad \textit{equivalently} \qquad \frac{1}{s} \;\le\; \frac{1}{g_K} \;\le\; \frac{1}{g_1}, \tag{14}$$

with every ladder increment $g_{k+1} - g_k \ge 0$. The hypothesis is used only for the Cholesky factorisation of Proposition 2.6 and is, on the surfaces of Section 8, certified per window.

Proof. Let $T = [t_1, \dots, t_h]$, so T is orthogonal and $\widehat{A}_h = T^\top A T$. Then $A^{-1} = T \widehat{A}_h^{-1} T^\top$, hence $t_1^\top A^{-1} t_1 = (\widehat{A}_h^{-1})_{11}$; and with $\mathcal{M}_h = \operatorname{diag}(\mu_1, \dots, \mu_h)$ and $G^{(h)} = \mathcal{M}_h^{-1/2} \widehat{A}_h \mathcal{M}_h^{-1/2}$ one has $\big((G^{(h)})^{-1}\big)_{11} = \mu_1 (\widehat{A}_h^{-1})_{11}$, i.e. $s = g_h$. Now apply (11): g_k is a partial sum of squares, so $g_1 \le g_K \le g_h = s$. □

2.4 Lag weights, the gauge law, and the boundary-free swap

Let $v \in \mathbb{R}^h$ and define the *lag weights* of v by

$$w_0 = \mathrm{Ac}_0(v), \qquad w_d = 2\,\mathrm{Ac}_d(v) - \mathrm{Cv}_{M-1-d}(v) \quad (1 \le d \le M-1), \qquad w_M := 0, \tag{15}$$

with Ac, Cv as in (1), and put $(\Delta w)_d = w_d - w_{d+1}$ and $\mathrm{TV} = \|\Delta w\|_1 = \sum_{d=0}^{M-1} |(\Delta w)_d|$. For an admissible x we write $w(x)$, $\mathrm{TV}(x)$ for the weights and total variation of the window vector $v = \sum_{k \le K} a_k t_k$, $a_k = x_k/\sqrt{\mu_k}$.

Lemma 2.9 (the weights represent the form; the gauge law)**.** *For every $v \in \mathbb{R}^h$ and every lag sequence c,*

$$v^{\mathsf{T}} A v = \sum_{d=0}^{M-1} c_d\, w_d, \qquad w_0 = \|v\|^2, \qquad \sum_{d=0}^{M-1} w_d = 0. \tag{16}$$

In particular, for an admissible x, $Q(x) = x^{\mathsf{T}} G x = v^{\mathsf{T}} A v = \sum_d c_d w_d(x)$.

Proof. By (2), $v^{\mathsf{T}} A v = \sum_{r,r'} v_r v_{r'} c_{|r-r'|} - \sum_{r,r'} v_r v_{r'} c_{M-1-r-r'}$. The first sum collects the lag $d = |r-r'|$ with multiplicity 1 for $d = 0$ and 2 for $d \geq 1$, i.e. it equals $c_0 \operatorname{Ac}_0(v) + 2 \sum_{d\geq 1} c_d \operatorname{Ac}_d(v)$. The second sum collects the index $d = M-1-r-r'$, and since $r, r' \leq h-1$ we have $r+r' \leq M-2$, so $d \geq 1$ throughout and the coefficient of c_d is $\operatorname{Cv}_{M-1-d}(v)$. This is the first identity of (16) with the weights (15), and it also shows $w_0 = \operatorname{Ac}_0(v) = \|v\|^2$: the reflection term cannot reach $d = 0$.

For the gauge law, sum (15). Over all $d \geq 0$, $\operatorname{Ac}_0(v) + 2\sum_{d\geq 1} \operatorname{Ac}_d(v) = \left(\sum_r v_r\right)^2$, and as d runs through $1, \ldots, M-1$ the index $j = M-1-d$ runs through $0, \ldots, M-2 = 2h-2$, so $\sum_{d\geq 1} \operatorname{Cv}_{M-1-d}(v) = \sum_{j=0}^{2h-2} \operatorname{Cv}_j(v) = \left(\sum_r v_r\right)^2$. The two contributions cancel. □

The gauge law has a consequence we use once: since $\sum_d w_d = 0$, the value $\sum_d c_d w_d$ is unchanged if a constant is added to every c_d. So the representation (16) determines the form only up to an additive normalisation of c, and any bound obtained from it may be optimised over that normalisation.

Lemma 2.10 (boundary-free summation by parts)**.** *With $C_d = \sum_{e\leq d} c_e$, $C_{-1} = 0$, and $w_M = 0$,*

$$\sum_{d=0}^{M-1} c_d\, w_d \;=\; \sum_{d=0}^{M-1} C_d\, (\Delta w)_d \tag{17}$$

with no boundary term at either end.

Proof. Substituting $c_d = C_d - C_{d-1}$,

$$\sum_{d=0}^{M-1} c_d w_d = \sum_{d=0}^{M-1} C_d w_d - \sum_{d=0}^{M-1} C_{d-1} w_d = \sum_{d=0}^{M-1} C_d w_d - \sum_{d=0}^{M-2} C_d w_{d+1} = \sum_{d=0}^{M-1} C_d (w_d - w_{d+1}),$$

where the second equality shifted the index and used $C_{-1} = 0$, and the third used $w_M = 0$ to absorb the last term. □

This is Abel's summation by parts [1]. Both boundary terms vanish for structural reasons and not by a choice: $C_{-1} = 0$ by definition of a partial sum, and $w_M = 0$ because both terms of (15) vanish at $d = M$.

3 The total-variation floor

Theorem 3.1 (the total-variation floor)**.** *Let $x \in \mathbb{R}^K$ be admissible, i.e. $x_1 = 1$, and let $w = w(x)$ be the lag weights of $v = \sum_{k\leq K} a_k t_k$, $a_k = x_k/\sqrt{\mu_k}$. Then*

$$\operatorname{TV}(x) \;\geq\; |w_0| \;=\; \|v\|^2 \;=\; \|a\|^2 \;\geq\; \frac{1}{\mu_1} \;=\; \frac{1}{4\sin^2(\pi/N)} \;\geq\; \frac{N^2}{4\pi^2}. \tag{18}$$

The bound involves no comb node, no taper and no truncation, and it holds for every lag sequence c — indeed the left- and right-hand sides do not depend on c at all.

Proof. Four steps.

(i) $w_0 = \|a\|^2$. By Lemma 2.9, $w_0 = \|v\|^2$; and $\|v\|^2 = \|a\|^2$ because $\{t_k\}$ is orthonormal (Lemma 2.2).

(ii) $\mathrm{TV} \ge |w_0|$. With $w_M = 0$ the increments telescope: $\sum_{d=0}^{M-1}(\Delta w)_d = w_0 - w_M = w_0$. Hence $|w_0| \le \sum_{d=0}^{M-1} |(\Delta w)_d| = \mathrm{TV}$.

(iii) $\|a\|^2 \ge 1/\mu_1$. Admissibility gives $a_1 = x_1/\sqrt{\mu_1} = 1/\sqrt{\mu_1}$, so $\|a\|^2 \ge a_1^2 = 1/\mu_1$.

(iv) $1/\mu_1 \ge N^2/(4\pi^2)$. This is (6), i.e. $\sin t \le t$. □

Three features of (18) are worth separating. It is a statement about the *normalisation*, not about the form: step (iii) is the only place where admissibility is used, and it is where the whole h^2 comes from. It is a statement about the *sector*: step (i) uses orthonormality of the parity modes and step (iv) uses their spectrum, both of which are the classical Kac–Murdock–Szegő data [13]. And it is *closed*: no constant is fitted, no window is excluded, and the right-hand side is an elementary function of h.

Corollary 3.2 (the price)**.** *For every admissible x,*

$$\frac{|Q(x)|}{\mathrm{TV}(x)} \;\le\; \mu_1\, |Q(x)|. \tag{19}$$

Consequently, if a family of admissible trial vectors has values bounded by $|Q(x)| \le U$ with U independent of h, the ratio it certifies is at most $U\mu_1 = O(h^{-2})$.

Proof. By Theorem 3.1, $\mathrm{TV}(x) \ge 1/\mu_1 > 0$, so $1/\,\mathrm{TV}(x) \le \mu_1$; multiply by $|Q(x)| \ge 0$. The consequence is $\mu_1 = 4\sin^2(\pi/N) \le 4\pi^2/N^2$. □

Corollary 3.2 is the load-bearing statement of the section. It says that the pair (value, total variation) cannot be improved by choosing a better trial vector: the ratio is pinned by the value alone, up to the spectral gap. Any route whose output is the ratio therefore pays a factor h^{-2} that is a property of the parity spectrum meeting the normalisation, and not a property of the comb.

Corollary 3.3 (what summation by parts can certify)**.** *For every admissible x,*

$$|Q(x)| \;\le\; \Big(\max_{0\le d\le M-1} |C_d|\Big)\; \mathrm{TV}(x), \qquad \textit{hence} \qquad \max_d |C_d| \;\ge\; \frac{|Q(x)|}{\mathrm{TV}(x)}. \tag{20}$$

Combined with (19)*, a family of bounded value certifies* $\max_d |C_d| \ge$ *a quantity that is $O(h^{-2})$, and nothing stronger. By the gauge law of Lemma 2.9 the estimate may be applied to $c + \gamma$ for any $\gamma \in \mathbb{R}$, so* $\max_d |C_d|$ *may be replaced by* $\inf_\gamma \max_d |C_d + \gamma(d+1)|$.

Proof. Lemma 2.10 and Hölder: $|\sum_d C_d(\Delta w)_d| \le \max_d |C_d| \cdot \|\Delta w\|_1$. The final sentence is the remark after Lemma 2.9, since replacing c_d by $c_d + \gamma$ leaves Q unchanged and replaces C_d by $C_d + \gamma(d+1)$. □

Corollary 3.4 (the floor at one coordinate)**.** *For every $v \in \mathbb{R}^h \setminus \{0\}$ with $t_1^{\mathsf{T}} v \ne 0$,*

$$\frac{\mathrm{TV}(v)}{(t_1^{\mathsf{T}} v)^2} \;\ge\; 1, \qquad \textit{because} \qquad \mathrm{TV}(v) \;\ge\; |w_0| = \|v\|^2 \;\ge\; (t_1^{\mathsf{T}} v)^2. \tag{21}$$

No normalisation is required: (21) *holds for every window vector, and for admissible x it reduces to* (18) *because* $(t_1^{\mathsf{T}} v)^2 = a_1^2 = 1/\mu_1$.

Proof. The first two relations are steps (i)–(ii) of Theorem 3.1, which used no admissibility; the third is Cauchy–Schwarz with $\|t_1\| = 1$. □

Remark 3.5 (negative control: the floor is a property of the normalisation)**.** Step (iii) is the only step that uses $x_1 = 1$, and it is indispensable: for $\lambda \neq 0$ the vector λx has $\mathrm{TV}(\lambda x) = \lambda^2\,\mathrm{TV}(x)$ (both Ac and Cv are quadratic in v, and v is linear in x), so $\mu_1\,\mathrm{TV}(\lambda x) = \lambda^2 \mu_1\,\mathrm{TV}(x)$ falls below 1 as soon as $|\lambda|$ is small enough. An inadmissible vector therefore undercuts the floor, and recognising it as inadmissible is the whole content of the observation. This is also the reason Theorem 4.1 below is a statement about rays.

Measurement 3.6 (sharpness of the floor)**.** On Surface II of Section 8 — 27 windows of the frame-A family, $h = 142$ to 1445, all with $\mathrm{cond}(G) \leq 10^{12}$ — the floor of Theorem 3.1 is tight to about one order of magnitude: the measured ratio is

$$\mu_1\ \mathrm{TV}(x) \in [5.00,\ 23.20] \tag{22}$$

at the optimiser of the declared trial family, while the floor itself takes the values $2.06 \cdot 10^3$ to $1.77 \cdot 10^5$, growing as $h^{+1.997}$ (a least-squares fit, reported for orientation and not used anywhere) against the closed h^2 of (6). The inadmissible control of Remark 3.5 returns $\mu_1\,\mathrm{TV} \approx 7 \cdot 10^{-4}$ at $\lambda = 10^{-2}$, i.e. it undercuts the floor by the predicted factor λ^2.

[Draft note: the declared caps of Surface II, and the definition of the trial family whose optimiser (22) *is evaluated at, are written out with Section 8.]*

We close the section by saying plainly what the floor does not know about. It does not know where the comb nodes are, or how many there are, or what their weights are; it does not know whether the trial vector was tapered, and it does not know at which K the mode family was truncated. It is a statement about $\{t_k\}$, about μ_1, and about the normalisation $x_1 = 1$. The next section shows that the last of these three cannot be traded away.

4 Gauge rigidity of the normalisation

Theorem 3.1 raises an obvious question: the floor is $1/\mu_1$ because the normalisation is imposed on the mode with eigenvalue μ_1, so could a different *entry spectrum* — a reweighted, shifted or otherwise reparametrised family — produce a flatter floor and hence a better ratio? The answer is no, and it is no for a structural reason.

Theorem 4.1 (the normalisation is a gauge)**.** *For $a \in \mathbb{R}^K$ let $v(a) = \sum_{k \leq K} a_k t_k$ and*

$$\widetilde{Q}(a) = v(a)^{\mathsf{T}} A v(a), \qquad \widetilde{\mathrm{TV}}(a) = \|\Delta w(v(a))\|_1. \tag{23}$$

Then:

(i) *$\widetilde{Q}$ and $\widetilde{\mathrm{TV}}$ are homogeneous of degree 2: $\widetilde{Q}(\lambda a) = \lambda^2 \widetilde{Q}(a)$ and $\widetilde{\mathrm{TV}}(\lambda a) = \lambda^2 \widetilde{\mathrm{TV}}(a)$ for all $\lambda \in \mathbb{R}$.*

(ii) *Hence the ratio $\varrho(a) = \widetilde{Q}(a)/\widetilde{\mathrm{TV}}(a)$ is constant on rays: it is a function of the direction of a alone.*

(iii) *Let $\mu' = (\mu'_1, \dots, \mu'_K)$ be* any *positive entry spectrum and let $x \mapsto a$, $a_k = x_k/\sqrt{\mu'_k}$, be the associated parametrisation. Then the constraint $x_1 = 1$ describes the affine slice $\{a : a_1 = 1/\sqrt{\mu'_1}\}$, every ray $\{\lambda a : \lambda > 0\}$ with $a_1 > 0$ meets that slice exactly once, and consequently*

$$\{\varrho(a) : a_k = x_k/\sqrt{\mu'_k},\ x_1 = 1\} \;=\; \{\varrho(a) : a \in \mathbb{R}^K,\ a_1 > 0\} \tag{24}$$

independently of μ'. *The entry normalisation is a gauge: the range of the ratio is the same in every sector.*

Proof. (i) $a \mapsto v(a)$ is linear, and by (1) both Ac_d and Cv_j are quadratic forms in v, so $w(v(\lambda a)) = \lambda^2 w(v(a))$ by (15), hence Δw and its ℓ^1 norm scale by λ^2; and $\widetilde{Q}$ is a quadratic form in a.

(ii) Immediate from (i), both numerator and denominator having the same degree.

(iii) The map $x \mapsto a$ is a linear bijection of $\mathbb{R}^K$ taking $\{x_1 = 1\}$ onto $\{a_1 = 1/\sqrt{\mu_1'}\}$. A ray $\{\lambda a : \lambda > 0\}$ with $a_1 > 0$ meets $\{a_1 = \theta\}$ exactly once for every $\theta > 0$, namely at $\lambda = \theta/a_1$. So the slice $\{a_1 = 1/\sqrt{\mu_1'}\}$ is a cross-section of the family of all rays with $a_1 > 0$, whatever the value of $\mu_1' > 0$; by (ii) the ratio takes the same set of values on the slice as on the family of rays, which is (24). The right-hand side does not mention μ'. □

Theorem 4.1 is a fact about degree-two homogeneous functionals, and we state it modestly for that reason. Its force is entirely in the consequence: since the accessible range of the ratio is a sector-independent set, and since Theorem 3.1 bounds that range in the parity parametrisation, no change of sector can enlarge it.

Corollary 4.2 (no sector change removes the h^2)**.** *Let μ' be any positive entry spectrum and let x' be admissible for it. Then*

$$\frac{|\widetilde{Q}(a')|}{\widetilde{\mathrm{TV}}(a')} \;\le\; \mu_1 \,\sup\{|\widetilde{Q}(a)| : a \textit{ in the parity slice}\}, \tag{25}$$

with $\mu_1 = 4\sin^2(\pi/N)$ the parity gap — i.e. the bound of Corollary 3.2 transfers verbatim, with the parity *gap, to every sector. In particular a family of trial vectors with h-bounded value in some sector certifies a ratio $O(h^{-2})$ in that sector as well.*

Proof. By (24) the value $\varrho(a')$ is attained by some a in the parity slice ($\mu' = \mu$), i.e. by some admissible x; apply Corollary 3.2 to that x. □

The next proposition is the quantitative form of Corollary 4.2 for the simplest reparametrisation, a rigid shift of the entry spectrum. It is a two-line computation, and we state it as a proposition rather than dressing it up: the point is not that the computation is hard but that the product it computes is *exactly* constant.

Proposition 4.3 (floor $\times$ transfer is conserved)**.** *Let $\tau \ge 0$ and let $\mu_k' = \mu_k + \tau$ be the shifted entry spectrum, so that admissibility in the shifted sector reads $a_1' = 1/\sqrt{\mu_1 + \tau}$. Then the shifted floor of Theorem 3.1 is $\widetilde{\mathrm{TV}}(a') \ge 1/(\mu_1 + \tau)$, the transfer factor relating the two normalisations is*

$$\theta = \frac{(a_1')^2}{a_1^2} = \frac{\mu_1}{\mu_1 + \tau}, \qquad \frac{1}{\theta} = 1 + \frac{\tau}{\mu_1}, \tag{26}$$

and

$$\underbrace{\frac{1}{\mu_1 + \tau}}_{\textit{shifted floor}} \cdot \underbrace{\Big(1 + \frac{\tau}{\mu_1}\Big)}_{\textit{transfer}} = \frac{1}{\mu_1 + \tau} \cdot \frac{\mu_1 + \tau}{\mu_1} = \frac{1}{\mu_1} \qquad \textit{identically, at every shift } \tau. \tag{27}$$

A shift does flatten the floor; it does not remove the h^2, which lives in the ratio and not in the floor.

Proof. The floor is Theorem 3.1 applied in the shifted parametrisation, whose step (iii) reads $\|a'\|^2 \ge (a_1')^2 = 1/(\mu_1 + \tau)$. For the transfer, the two admissibility constraints differ only in the value of the first coefficient, $a_1 = 1/\sqrt{\mu_1}$ against $a_1' = 1/\sqrt{\mu_1 + \tau}$; by Theorem 4.1(i) a value computed at a' is converted to the corresponding value on the same ray in the unshifted slice by the factor $(a_1/a_1')^2 = 1/\theta$. The displayed product is then (27). □

Proposition 4.4 (the null mode, and the other sector). *Let $c^L = (2, -1, 0, \dots)$ be as in Lemma 2.2.*

(i) *The parity family $\mu_k = 4\sin^2(k\pi/N)$, $k = 1, \dots, h$, is the restriction to $k \geq 1$ of the sine family $\{4\sin^2(k\pi/N)\}_{k\geq 0}$, whose member at $k = 0$ is $\mu_0 = 0$.*

(ii) *The parametrisation $a_k = x_k/\sqrt{\mu'_k}$ of Theorem 4.1*(iii) *requires $\mu'_k > 0$. If a mode family containing a null mode is used and the normalisation is imposed on that mode, the constraint has* no *representative a of finite norm; if the normalisation is imposed elsewhere, step (iii) of Theorem 3.1 yields the floor of the normalised mode and the null mode contributes nothing. In neither case is a flatter admissible floor obtained.*

(iii) *Write $\Pi_+ e_r = \frac{1}{\sqrt{2}}(e_r + e_{M-1-r})$ for the reflection-even isometry. The complementary compression $\Pi_+^\top T_M(c^L)\Pi_+$ has spectrum $\{4\sin^2((2k+1)\pi/(2N))\}_{k=0}^{h-1}$, whose minimum $4\sin^2(\pi/(2N))$ is* strictly smaller *than μ_1. Hence no reflection sector of the second-difference form has a lowest eigenvalue exceeding μ_1.*

Proof. (i) is the formula of Lemma 2.2 read at $k = 0$.

(ii) is the definition of the parametrisation together with step (iii) of Theorem 3.1, which bounds $\|a\|^2$ below by the square of the *normalised* coordinate only.

(iii) For c^L the even compression is $(\Pi_+^\top T_M(c^L)\Pi_+)_{rr'} = c^L_{|r-r'|} + c^L_{M-1-r-r'}$, which by the computation in the proof of Lemma 2.2 is tridiagonal $2I - E - E^\top$ with corner entry $2 - 1 = 1$; as a recurrence this is $u_{-1} = 0$ and $u_h = +u_{h-1}$. With $u_r = \sin(\omega(r+1))$ the inner condition reads $\sin(\omega(h+1)) - \sin(\omega h) = 2\cos(\frac{\omega N}{2})\sin(\frac{\omega}{2}) = 0$, hence $\omega N/2 = (k + \frac{1}{2})\pi$ and $\mu = 4\sin^2((2k+1)\pi/(2N))$ for $k = 0, \dots, h-1$, which are h distinct values, hence all. The minimum is at $k = 0$ and $4\sin^2(\pi/(2N)) < 4\sin^2(\pi/N) = \mu_1$ because sin is increasing on $[0, \pi/2]$. □

Part (iii) is the precise sense in which the parity compression is the best of the two reflection sectors for this purpose, and part (ii) is the sense in which enlarging the mode family removes the statement instead of improving it. Together with Theorem 4.1 and Proposition 4.3 this closes the sector question: reweighting, shifting, or enlarging the entry spectrum changes neither the accessible range of the ratio nor the conserved product.

Measurement 4.5 (the sector battery). Theorem 4.1 and Proposition 4.3 are proved statements; the following is their machine verification on a declared battery, reported because it also checks the implementation of the transport map. Over 35 weight families $\times$ 6 taper widths $\times$ 27 windows of Surface II the ratio ϱ is reproduced across sectors with residual at most $8.8 \cdot 10^{-10}$ against a declared round-off horizon of $2.6 \cdot 10^{-8}$, and the product (27) is reproduced as an exact identity at every shift of the declared shift list. Transporting the x-coordinates instead of the ray — the wrong map — moves the ratio by a large factor on the spread-out weight families, which is the mutation control of the battery.

5 The two-dimensional reduction

[Draft note: Sections 5–9 carry their statements in final form together with proof sketches. The full write-ups, the priced table of Section 7 and the measurement tables of Section 8 are completed in a later pass; the type of each statement is recorded in Table 1 and nothing is upgraded in the meantime.]

Sections 3 and 4 priced the trial-vector route. This section identifies what is left: at $K = 2$ the optimal trial vector is available in closed form, so the entire question collapses onto a single scalar, the squared correlation of the two lowest parity modes in the metric of A.

Proposition 5.1 (the closed vector is exact at $K = 2$). *Assume $G_2 \succ 0$ and put $r_{12}^2 = G_{12}^2/(G_{11}G_{22}) = \widehat{a}_{12}^2/(\widehat{a}_{11}\widehat{a}_{22})$. Then $u = (1, -G_{21}/G_{22})$ attains the constrained minimum of Lemma 2.5 at $k = 2$:*

$$u^\top G_2 u = \frac{1}{g_2} = G_{11}(1 - r_{12}^2), \qquad G_{11}\, g_2 = \frac{1}{1 - r_{12}^2} \quad \textit{identically.} \tag{28}$$

Proof sketch. Lemma 2.4 at $k = 2$ gives $1/g_2 = G_{11} - G_{12}^2/G_{22} = G_{11}(1 - r_{12}^2)$, and Lemma 2.5 identifies the minimiser as $G_2^{-1}e_1/(G_2^{-1}e_1)_1 = (1, -G_{21}/G_{22})$. The second identity is the first divided by $g_1 = 1/G_{11}$, i.e. the case $k = 2$ of (13). □

Proposition 5.2 (pivot identity). *Assume $G_k \succ 0$ and let $x^\star$ be the minimiser of Lemma 2.5. Then for every δ with $\delta_1 = 0$,*

$$(x^\star + \delta)^\top G_k (x^\star + \delta) = \frac{1}{g_k} + \delta^\top G_k \delta \qquad \textit{exactly,} \tag{29}$$

so the relative excess $\rho = g_k\, \delta^\top G_k \delta$ of any explicit candidate is an exact number and never an estimate.

Proof sketch. $G_k x^\star = e_1/g_k$ and $\delta_1 = 0$ annihilate the cross term; this is the computation already carried out in the proof of Lemma 2.5. □

Proposition 5.3 (the entrywise perturbation, and one cancellation ratio). *Let $x^\star$ be as above, $\sigma = \operatorname{sgn}(x^\star)$, $\Theta_k = |x^\star|^\top |G_k|\, |x^\star|$, and for $\varepsilon > 0$ let E be the sign-aligned entrywise perturbation $E_{ij} = \varepsilon |G_{ij}| \sigma_i \sigma_j$. Then*

$$\bigl|(x^\star)^\top E x^\star\bigr| = \varepsilon\, \Theta_k \qquad \textit{exactly,} \tag{30}$$

so the entry threshold at which the perturbation exhausts the value is $\varepsilon_{\mathrm{ent}}(k) = \rho\,(1/g_k)/\Theta_k$ with ρ of (29). At $k = 2$ the ratio is closed:

$$\Theta_2 = G_{11}(1 + 3r_{12}^2), \qquad \frac{1/g_2}{\Theta_2} = \frac{1 - r_{12}^2}{1 + 3r_{12}^2}, \tag{31}$$

and $1 + 3r_{12}^2 \to 4$ as $r_{12}^2 \to 1$.

Proof sketch. (30) is the definition of E: the signs align every term, so the sum of $|x_i^\star||x_j^\star||G_{ij}|$ is attained without cancellation. At $k = 2$, $|x^\star| = (1, |G_{12}|/G_{22})$, whence $\Theta_2 = G_{11} + 2G_{12}^2/G_{22} + G_{12}^2/G_{22} = G_{11}(1 + 3r_{12}^2)$; divide by (28). □

Remark 5.4. Proposition 5.3 is the reason the three natural readings of the same obstruction — a determinant ratio, the scalar $1 - r_{12}^2$, and an entrywise perturbation threshold — are one object rather than three: at $k = 2$ they differ by the explicit factor $1 + 3r_{12}^2 \in [1, 4]$. We state the two formulae and leave the reading to the reader; "three dresses of one inequality" is an observation about identities, not a theorem.

Lemma 5.5 (Wronskian telescope). *Let $u = t_1$, $v = t_2$ be the two lowest parity modes of Lemma 2.2, extended by $u_{-1} = v_{-1} = 0$ and $u_h = -u_{h-1}$, $v_h = -v_{h-1}$, and put $W_r = u_{r+1}v_r - u_r v_{r+1}$. Then*

$$(\mu_1 - \mu_2)\, u_r v_r = W_{r-1} - W_r \quad (0 \le r \le h-1), \qquad W_{-1} = W_{h-1} = 0, \tag{32}$$

and summing (32) re-derives $t_1^\top t_2 = 0$.

Proof sketch. $W_{r-1} - W_r = u_r(v_{r-1} + v_{r+1}) - v_r(u_{r-1} + u_{r+1})$, and the recurrences $u_{r-1} + u_{r+1} = (2 - \mu_1)u_r$, $v_{r-1} + v_{r+1} = (2 - \mu_2)v_r$ give (32). $W_{-1} = 0$ is the Dirichlet end; $W_{h-1} = 0$ is the odd reflection $u_h = -u_{h-1}$, $v_h = -v_{h-1}$, i.e. exactly the corner entry 3 of L_P. □

Lemma 5.6 (maximal minors carry no arithmetic). *With $u = t_1/\sqrt{\mu_1}$, $v = t_2/\sqrt{\mu_2}$ and $\mathcal{W}_{rr'} = u_r v_{r'} - u_{r'} v_r$,*

$$\tfrac{1}{2}\sum_{r,r'} \mathcal{W}_{rr'}^2 = \|u\|^2\|v\|^2 - \langle u, v\rangle^2 = \frac{1}{\mu_1\mu_2} \qquad \textit{exactly.} \tag{33}$$

Proof sketch. Lagrange's identity [15] gives the first equality; the second is orthonormality of t_1, t_2 (Lemma 2.2). □

Lemma 5.7 (the determinant reading, free of positivity). *With $\nu_1 \geq \nu_2$ the eigenvalues of $\widehat{A}_2$,*

$$1 - r_{12}^2 = \frac{\det \widehat{A}_2}{\widehat{a}_{11}\widehat{a}_{22}} = \frac{\nu_1\nu_2}{\widehat{a}_{11}\widehat{a}_{22}} = \frac{\det G_2}{G_{11}G_{22}}, \tag{34}$$

and no positivity of A, of $\widehat{A}_2$ or of G is used.

Proof sketch. $\det \widehat{A}_2 = \widehat{a}_{11}\widehat{a}_{22} - \widehat{a}_{12}^2 = \nu_1\nu_2$; divide by $\widehat{a}_{11}\widehat{a}_{22}$. The last expression is the same computation after the diagonal congruence (7), which cancels by Lemma 2.7. □

Lemma 5.7 is load-bearing in a negative sense, and we say so plainly: it is what makes the whole reduction free of any positivity input about A. Whether A is positive definite on the parity sector is, on the surfaces of Section 8, a numerical fact about finitely many finite matrices; no statement of this paper is routed through a uniform version of it.

Lemma 5.8 (an unconditional bound for the large eigenvalue). $\nu_1 \leq \max(\widehat{a}_{11}, \widehat{a}_{22}) + |\widehat{a}_{12}|$, *closed in the three entries and uniform in the window.*

Proof sketch. Gershgorin's theorem [8] for the symmetric 2×2 matrix $\widehat{A}_2$. □

Proposition 5.9 (the determinant returns: the $t^\star$ collapse). *Assume $\widehat{a}_{11}, \widehat{a}_{22} > 0$ and consider the one-parameter family $u(t) = (t, -1)$. At $t^\star = \sqrt{\widehat{a}_{22}/\widehat{a}_{11}}$ the Rayleigh quotient of $\widehat{A}_2$ is*

$$R(t^\star) = \frac{u(t^\star)^{\mathsf{T}}\widehat{A}_2 u(t^\star)}{\|u(t^\star)\|^2} = \frac{2\sqrt{\widehat{a}_{11}\widehat{a}_{22}}\,\det \widehat{A}_2}{(\widehat{a}_{11} + \widehat{a}_{22})\big(\sqrt{\widehat{a}_{11}\widehat{a}_{22}} + \widehat{a}_{12}\big)} \qquad \textit{identically,} \tag{35}$$

so the only member of the family that meets the threshold does so by reintroducing $\det \widehat{A}_2$. *If in addition $\widehat{A}_2 \succ 0$, then*

$$\nu_2 \;\leq\; \frac{2\det \widehat{A}_2}{\widehat{a}_{11} + \widehat{a}_{22}} \;=\; \frac{2\nu_1\nu_2}{\nu_1 + \nu_2} \;\leq\; 2\nu_2, \tag{36}$$

i.e. the harmonic mean of the two eigenvalues bounds ν_2 from above and is tight within a factor 2.

Proof sketch. For (35): with $\tau = t^\star$ the numerator is $\tau^2\widehat{a}_{11} - 2\tau\widehat{a}_{12} + \widehat{a}_{22} = 2(\widehat{a}_{22} - \tau\widehat{a}_{12}) = 2\tau\big(\sqrt{\widehat{a}_{11}\widehat{a}_{22}} - \widehat{a}_{12}\big)$ and the denominator is $\tau^2 + 1 = (\widehat{a}_{11} + \widehat{a}_{22})/\widehat{a}_{11}$; multiply numerator and denominator by $\sqrt{\widehat{a}_{11}\widehat{a}_{22}} + \widehat{a}_{12}$ and use $\widehat{a}_{11}\widehat{a}_{22} - \widehat{a}_{12}^2 = \det \widehat{A}_2$. For (36), substitute $\operatorname{tr} \widehat{A}_2 = \nu_1 + \nu_2$ and $\det \widehat{A}_2 = \nu_1\nu_2$; the two inequalities are $\nu_2 \leq \nu_1$ and $\nu_1 + \nu_2 \geq \nu_1$ respectively, both of which use $\nu_2 > 0$. Without positivity (36) is false — if $\nu_1 > 0 > \nu_2$ and $\operatorname{tr} > 0$ the right-hand side is negative — which is why the hypothesis is named. □

Remark 5.10. Propositions 5.1–5.9 are exact reformulations of one another, and none of them lowers the accuracy demanded of $\det \widehat{A}_2$: the determinant is reintroduced by (35), the scalar $1 - r_{12}^2$ is the determinant normalised by (34), and the entry threshold of (31) differs from it by the bounded factor $1 + 3r_{12}^2$. We record this as an observation about the identities, not as a theorem; the statement "every exact reformulation costs the same precision" is a reading of a family of identities together with measured accuracy demands, and it belongs in Section 7 in that form.

6 The exact bilinear form and the rank classification

This section writes $\widehat{A}_2$ out in terms of the comb and identifies the exact algebraic shape of the resulting object. Throughout the section the comb nodes are indexed by ℓ, so that i, j stay free for matrix indices.

Proposition 6.1 (exact split). *Let D be admissible, $D < \min_\ell u_\ell$, and for a node u_ℓ let $m_\ell \in \mathbb{N}$ and $\theta_\ell \in [0,1)$ be defined by $u_\ell/D = m_\ell + \theta_\ell$. Put $Y^{(m)} = \Pi^\top T_M(\delta_m)\Pi$ with δ_m the lag sequence supported at the single lag m, and $Y^{(m)} := 0$ for $m \geq M$. Then, with $B_{ij} = t_i^\top A_{\mathrm{arch}} t_j$ and*

$$X_{ij}^{(\ell)} = \tfrac{1}{2}\Big[(1-\theta_\ell)\, t_i^\top Y^{(m_\ell)} t_j + \theta_\ell\, t_i^\top Y^{(m_\ell+1)} t_j\Big], \qquad S = \sum_\ell \varrho_\ell X^{(\ell)} \qquad (i, j \in \{1, 2\}), \tag{37}$$

one has $\widehat{A}_2 = B - S$ exactly, and each of S_{11}, S_{22}, S_{12} is a linear *functional of the weight vector $(\varrho_\ell)_\ell$.*

Proof sketch. Linearity of $c \mapsto A \mapsto \widehat{A}$ applied to (3) and (4). For the two-point read: $S_D(mD, u_\ell) = \frac{1}{2}[\Delta_D(mD - u_\ell) + \Delta_D(mD + u_\ell)]$; the reflected hat vanishes for every $m \geq 0$ because $mD + u_\ell \geq u_\ell > D$, and the direct hat is nonzero only for $m \in \{m_\ell, m_\ell + 1\}$, where the unit-height triangle Δ_D takes the values $1 - \theta_\ell$ and θ_ℓ respectively. The factor $\frac{1}{2}$ is the symmetrisation, and a lag index $\geq M$ falls outside the window and contributes nothing. $\square$

Proposition 6.2 (polarisation of the determinant). *With $D(P, Q) = P_{11}Q_{22} + P_{22}Q_{11} - 2P_{12}Q_{12}$,*

$$\det \widehat{A}_2 = \det B - D(B, S) + \det S, \tag{38}$$

where the three terms are respectively archimedean–archimedean, linear in the comb weights, and bilinear in them. Moreover

$$\det S = \sum_{\ell, \ell'} \varrho_\ell \varrho_{\ell'}\, \mathcal{K}(\ell, \ell'), \qquad \mathcal{K}(\ell, \ell') = \tfrac{1}{2} D(X^{(\ell)}, X^{(\ell')}), \tag{39}$$

with $X^{(\ell)}$ the 2×2 matrix of (37).

Proof sketch. Both are instances of Theorem 6.3(i) below, applied to $\widehat{A}_2 = B - S$ and to $S = \sum_\ell \varrho_\ell X^{(\ell)}$ respectively; the second uses bilinearity of D and $\det S = \frac{1}{2} D(S, S)$ [15, 2]. $\square$

Theorem 6.3 (rank three). *Identify a symmetric 2×2 matrix X with its coordinate vector $y(X) = (X_{11}, X_{22}, X_{12}) \in \mathbb{R}^3$ and let*

$$J = \begin{pmatrix} 0 & 1 & 0 \\ 1 & 0 & 0 \\ 0 & 0 & -2 \end{pmatrix}. \tag{40}$$

Then:

(i) $\det X = \frac{1}{2}\, y(X)^\top J y(X)$ *and* $D(P, Q) = y(P)^\top J y(Q)$*; in particular D is the polarisation of* det *and* $\det(P - Q) = \det P - D(P, Q) + \det Q$.

(ii) *J has eigenvalues $\{1, -1, -2\}$: it has* ***rank three*** *and* ***signature*** $(1, 2)$.

(iii) *Let $P \in \mathbb{R}^{3 \times n}$ have columns $y(X^{(\ell)})$, $\ell = 1, \ldots, n$. Then the kernel of* (39) *is $\mathcal{K} = \frac{1}{2} P^\top J P$, so*

$$\operatorname{rank} \mathcal{K} \;\leq\; 3 \qquad \textit{for every window, every } h \textit{ and every truncation,} \tag{41}$$

and the nonzero spectrum of $P^\top J P$ equals the nonzero spectrum of $J P P^\top$, a 3×3 matrix.

(iv) *Consequently* $\det S = S_{11}S_{22} - S_{12}^2$*: the comb–comb term of* (38) *is a quadratic polynomial in exactly* three *linear functionals of the comb weights, and depends on the weight vector* (ϱ_ℓ) *through nothing else.*

Proof sketch. (i) Both sides of the first identity are $X_{11}X_{22} - X_{12}^2$; reading off the bilinear form gives $y(P)^\top J y(Q) = P_{11}Q_{22} + P_{22}Q_{11} - 2P_{12}Q_{12} = D(P,Q)$, and expanding $\frac{1}{2}(y(P) - y(Q))^\top J(y(P) - y(Q))$ gives the third relation. (ii) The upper-left 2×2 block $\left(\begin{smallmatrix} 0 & 1 \\ 1 & 0 \end{smallmatrix}\right)$ has eigenvalues ± 1, and the remaining entry is -2. (iii) $\mathcal{K}(\ell, \ell') = \frac{1}{2}y(X^{(\ell)})^\top J y(X^{(\ell')})$ is the (ℓ, ℓ') entry of $\frac{1}{2}P^\top JP$, whose rank is at most $\operatorname{rank} J = 3$; the statement about spectra is the standard fact $\operatorname{spec}(UV) \setminus \{0\} = \operatorname{spec}(VU) \setminus \{0\}$ with $U = P^\top$, $V = JP$. (iv) is (i) applied to S, whose coordinates are the three linear functionals of Proposition 6.1. $\square$

Part (ii) will strike a reader as elementary — the determinant of a 2×2 matrix is visibly a quadratic form in its three entries — and it is. What the theorem is for is part (iv) together with the following corollary, which is where the elementary observation has teeth.

Corollary 6.4 (what a majorant of the comb–comb term can see)**.** *Let* $\mathcal{N}$ *be any norm on the weight vectors and suppose* $|\det S| \le \Phi(\mathcal{N}(\varrho))$ *for all admissible weight vectors, with* Φ *nondecreasing. Then necessarily*

$$\Phi(t) \;\ge\; \sup\{|y_1 y_2 - y_3^2| : \; y \in \mathbb{R}^3, \; y = (S_{11}, S_{22}, S_{12}) \textit{ for some } \varrho \textit{ with } \mathcal{N}(\varrho) \le t\}. \tag{42}$$

In particular, no estimate that reaches $\det S$ *through a norm of* ϱ *can distinguish two weight vectors with the same image* (S_{11}, S_{22}, S_{12})*, and any cancellation between the three terms of* (38) *is invisible to it.*

Proof sketch. Immediate from Theorem 6.3(iv): $\det S$ is a function of the image point only, so a bound valid for all ϱ in a norm ball must dominate the supremum over the image of that ball. $\square$

Measurement 6.5 (Type II blocks: a measured rank collapse)**.** The closed weights of Proposition 6.1 see a node only through its position u_ℓ, i.e. for the prime-power instance only through $\log n_\ell$; under a factorisation $n = bd$ that is $\log b + \log d$, and a bounded-frequency function of a sum is a finite-rank kernel. On the declared surface the Type II blocks of Vaughan's identity [21] built from these weights are measured to have 2 singular values carrying 99.9% of the mass, against 153 for a generic kernel of the same size, over the scanned range of the decomposition parameter; the four-term identity for Λ is verified coefficientwise first.

[Draft note: This row is measured *and stays measured. "Effective rank* $O(1)$*" is not a theorem. The provable substitute* — if *the weight kernel admits an* Ω*-band expansion in* $\log b + \log d$ *with declared truncation error* η*, then the block has* η*-rank at most* $2\Omega + 1$ — *requires a decay estimate for the closed weight kernel in* $\log n$ *that is not written in this revision. Bandwidth* $\Rightarrow$ η*-rank lemma: open writing task.]*

Measurement 6.6 (the reference window: a three-term cancellation)**.** On the reference window $h = 540$, with the comb truncated at the node cap $2.465 \cdot 10^4$, the three terms of (38) are

$$\det B = 6.18, \qquad -D(B,S) = -198.8, \qquad \det S = 192.6, \qquad \text{sum} = 4.22 \cdot 10^{-3}, \tag{43}$$

i.e. five orders of magnitude of cancellation between three pieces of size ≈ 200; the target value at that window is $1.10 \cdot 10^{-5}$.

[Draft note: the window's caps and the definition of the target are written out with Section 8.]

7 A measured obstruction, and an open problem

Problem 7.1 (three equivalent forms)**.** Is there, for each $\varepsilon > 0$, a constant C_ε such that *uniformly in the window index*,

$$1 - r_{12}^2 = \frac{\det \widehat{A}_2}{\widehat{a}_{11}\widehat{a}_{22}} \;\le\; C_\varepsilon\, h^{-3+\varepsilon}\,? \tag{44}$$

Equivalently, by Lemma 5.7 and Lemma 5.6:

(a) *Row collinearity.* Do the two rows of $\widehat{A}_2$ — two explicit finite sums against the closed weights of Proposition 6.1 — become collinear at rate $h^{-3+\varepsilon}$, the sine of the angle between them being $|\det \widehat{A}_2|/(\|\mathrm{row}_1\|\|\mathrm{row}_2\|)$?

(b) *Determinant form.* Does the normalised determinant $\det \widehat{A}_2/(\widehat{a}_{11}\widehat{a}_{22})$ satisfy (44)?

(c) *Joint near-degeneracy.* Does the triple (S_{11}, S_{22}, S_{12}) of Theorem 6.3(iv) approach the archimedean block B jointly, at relative precision sharpening like h^{-3}, in the sense that the three terms of (38) cancel to that order?

The problem is open. It is a question about a finite bilinear form; it is not formulated as, and must not be read as, an approach to, a weakening of, or a consequence of any conjecture, and no comparison of strengths is made anywhere in this paper.

Measurement 7.2 (measured rates)**.** On the declared surfaces the left-hand side of (44) is measured to fall as a power of h with fitted exponent between -2.7 and -2.95 depending on the surface, and the sine-of-angle reading of Problem 7.1(a) falls at the same rate to within the fit uncertainty. Fitted exponents are fits.

[Draft note: the surfaces, the window counts, the caps, and the exponents with their jackknife bands are written out in the next pass; nothing here is used as a bound.]

Measurement 7.3 (priced routes: a table of what the unconditional toolbox reaches)**.** Each row of the table below is an unconditional bound, evaluated on a declared surface, together with the exponent it certifies; the bound is proved, the exponent is measured, and the two are typed separately. The routes are: Cauchy–Schwarz on $\widehat{A}_2$; the large sieve and duality applied to $\det S$ [16]; Montgomery–Vaughan applied to the three linear functionals of Theorem 6.3(iv); the exact split $|\det B| + |D(B,S)| + |\det S|$ of (38); and Vaughan's Type I + Type II decomposition [21]. Alongside them sit the structural no-gos: the symbol infimum (negative on every window of the declared surface, so no symbol-positivity argument can deliver a floor); perturbation theory around the parity bottom mode, which bounds at the scale of the matrix [14]; band-local domination; and the ℓ^1 mass of the kernel $\mathcal{K}$ against its signed value.

[Draft note: Placeholder for the priced table. Each row needs its route's hypotheses spelled out and its exponent re-read from a declared surface with caps; until that is written, no exponent is quoted here. Corollary 6.4 is the structural reason the norm-based rows lose.]

Measurement 7.4 (the density curve)**.** The gap between the rate delivered by the best route of Measurement 7.3 and the rate $h^{-3+\varepsilon}$ of Problem 7.1, measured against the density of comb nodes per lag cell, falls monotonically over three decades of density; the densest reachable bin is consistent with zero. A true zero, a power-law approach and a low plateau are *undecidable* under the finite sieve that the declared caps permit, and this undecidability is part of the claim, not a caveat attached to it.

[Draft note: Placeholder for the density curve: the six binned values with their standard errors, the caps that make the dense corner reachable, the jackknife correction to the scatter, and the explicit undecidability sentence.]

Measurement 7.5 (placement discriminators)**.** Two controlled experiments locate where the arithmetic actually sits. Replacing the node positions by sorted uniform samples at the *same* weight multiset leaves the rank-three collapse of Theorem 6.3 and every unconditional bound of Measurement 7.3 numerically almost unchanged, while the *value* of $1 - r_{12}^2$ moves by orders of magnitude; and displacing a few nodes by a small fraction of a lag cell moves log of the ratio linearly over several declared decades, with a first-harmonic prediction that under-predicts by a large factor — the response is not smooth in the placement. The theorems of this paper are therefore provably not where the arithmetic is, and the equidistribution of the relevant cell phases is a classical Weyl statement [22].

[Draft note: Placeholder: the scramble factors, the interventional slope, the declared decades and the positive-definiteness threshold.]

Remark 7.6 (what is not claimed)**.** Problem 7.1 is open. Nothing in this paper bounds it, and nothing in this paper is conditional on it. In particular: no statement here is an arithmetic statement; the positivity of A on the parity sector is used only where it is named as a hypothesis, and only as a per-window certificate; and the comparison of a demanded precision against the strength of any conjecture is deliberately absent from the whole paper.

8 Numerical verification

[Draft note: Section skeleton. The floating-point convention, the two declared surfaces with their caps, and the certificate inventory are written out in the next pass; the cross-reference table below is complete as it stands.]

8.1 Floating-point convention and the two surfaces

Positivity statements are completed Cholesky factorisations relative to a declared floor in the sense of Wilkinson and Higham [23, 12]: with $u = 2^{-53}$ the unit roundoff and $c_h = (h+1)/(1-(h+1)u)$, a completed factorisation of $X - \delta I$ with $\delta = c_h u \|X\|_\infty$ certifies $\lambda_{\min}(X) > 0$, and the cheap certified bound $\|X\|_2 \leq \|X\|_\infty$ is used for the norm [8]. Quantities obtained from an eigenvalue routine are labelled *measured* and never used as certificates. Diagonally dominant comparisons, where used, are Ostrowski's [17].

Two surfaces are declared and never mixed. *Surface I* (exact, small): the deepest nodes admitting a window inside a cap on the lag index, on which every identity of Sections 2–6 is recomputed from scratch. *Surface II* (measurement): the frame family with $h = 142$ to 1445 and $\mathrm{cond}(G) \leq 10^{12}$, on which Measurements 3.6, 4.5, 6.6 and 7.2 are taken.

8.2 Cross-reference table

Table 3: Every numbered statement and the check that recomputes it. Module identifiers refer to the verification suite accompanying this work; item numbers are those of the module documentation.

statement	check	what the check recomputes
Lem. 2.2 (parity spectrum)	`v555–v559`, parity control	$L_P t_k = \mu_k t_k$ with an orthonormal basis; $\Pi^\mathsf{T} T_M(c^L)\Pi = L_P$ at zero tolerance

continued on the next page

statement	check	what the check recomputes
Lem. 2.9 (gauge law)	`v555` item (4); `v559` link 3	$\sum_d w_d = 0$; $\sum_d c_d w_d = x^\mathsf{T} G x$
Lem. 2.10 (Abel swap)	`v555` item (4); `v559` link 9	the boundary-free identity at levels $k = 1, 2, 3$
Prop. 2.6 (three ladder forms)	`v557` item (1) I1–I3; `v559` link 12	prefix from one Cholesky; minor form; regression form
Lem. 2.7 (diagonal invariance)	`v557` item (2)	invariance of g_K/g_1 under $G \mapsto \Lambda G \Lambda$
Prop. 2.8 (ladder direction)	`v556` item (5); `v559` link 4	$s \geq g_K \geq g_1$ with strictly positive increments
Thm. 3.1 (TV floor)	`v555` item (2), steps T1–T3	$w_0 = \|a\|^2$; $\mathrm{TV} \geq \lvert w_0 \rvert$; $\mu_1 \, \mathrm{TV} \geq 1$
Cor. 3.3 (Abel/Hölder)	`v555` item (4); `v559` link 9	$\lvert Q \rvert \leq \mathrm{TV} \max_d \lvert C_d \rvert$
Cor. 3.4 (floor at one coordinate)	`v556` item (4b)	$\mathrm{TV} \geq \|v\|^2 \geq (t_1^\mathsf{T} v)^2$
Rem. 3.5 (negative control)	`v555` item (2), control	an inadmissible vector undercuts the floor
Thm. 4.1 (gauge)	`v556` item (1); `v559` link 10	degree-two homogeneity; ratio equal across sectors
Prop. 4.3 (conservation)	`v556` item (2)	floor $\times$ transfer $= 1/\mu_1$ at every shift
Prop. 5.1 ($K = 2$ exact)	`v557` item (3); `v559` link 13	the closed vector attains $1/g_2$; $G_{11} g_2 = 1/(1 - r_{12}^2)$
Prop. 5.2, 5.3 (pivot, entrywise)	`v557` item (4)	the pivot identity; $\lvert x^{\star\mathsf{T}} E x^\star \rvert = \varepsilon \Theta_K$
Lem. 2.5 (trial direction)	`v557` item (5); `v559` links 4, 7	every admissible trial is a lower bound for $1/g_K$
Lem. 5.5, 5.6 (Wronskian, minors)	`v558` item (1) I3	the telescope with $W_{-1} = W_{h-1} = 0$; $\frac{1}{2} \sum \mathcal{W}^2 = 1/(\mu_1 \mu_2)$
Lem. 5.7 (determinant reading)	`v558` items (2), (6); `v559` link 14	the three equal expressions, no positivity used
Lem. 5.8 (Gershgorin)	`v558` item (2)	$\nu_1 \leq \max(\widehat{a}_{11}, \widehat{a}_{22}) + \lvert \widehat{a}_{12} \rvert$
Prop. 5.9 ($t^\star$ collapse)	`v558` item (3)	the closed form (35); the harmonic-mean bound
Prop. 6.1, 6.2 (split, polarisation)	`v558` item (4) TH1–TH3; `v559` link 15	$\widehat{A}_2 = B - S$; (38); the kernel $\mathcal{K}$
Thm. 6.3 (rank three)	`v558` item (5) TH4; `v559` link 15	J, its spectrum, $\operatorname{rank} \mathcal{K} \leq 3$
Meas. 6.5 (Type II)	`v558` item (5) TH5	singular-value profile against a generic kernel
Cor. 6.4 (majorants)	`v558` items (4), (6)	the wedge reading; the scramble discriminator
Prob. 7.1 (the open problem)	`v559` link 16	the *shape* is certified per window; the rate and the quantifier stay open
Meas. 7.3 (priced routes)	`v559` NG1–NG8	each classified route exhibited failing as classified

8.3 Reproducibility

[Draft note: Placeholder: the scripts, their runtimes, the number of checks, and the statement that every factorised matrix stays under the declared cap.]

9 Limitations, and what is not claimed

(L1) **No uniformity in the window index.** Every statement typed *certified per window* in Table 1 is verified window by window on a declared finite surface, and no quantifier over windows is claimed. In particular the hypothesis of Proposition 2.8 and the positivity hypothesis of (36) are certificates, not theorems.

(L2) **Positivity facts are facts about finitely many finite matrices.** That A is positive definite on the parity sector on the tested windows is a numerical statement with a declared floor. It is not routed through, compared with, or used to suggest anything about any criterion; Lemma 5.7 exists precisely so that the reduction never needs it.

(L3) **Fits are fits.** Every exponent in Measurements 3.6, 7.2 and 7.4 is a least-squares fit on a declared surface, reported for orientation. No fit is used as a bound anywhere.

(L4) **The dense corner is exhausted only up to the declared caps.** Measurement 7.4 states its own undecidability, and that statement is part of the claim.

(L5) **One statement did not convert.** Measurement 6.5 stays measured: the bandwidth $\Rightarrow$ η-rank lemma that would make it a theorem is not written, and until it is, "effective rank $O(1)$" is a measurement and is labelled one.

(L6) **The comb's arithmetic is not used by any theorem.** Remark 2.1 is meant literally: replacing the prime-power instance by an arbitrary finite comb changes no proof in Sections 2–6. What changes is the *value* of the three functionals of Theorem 6.3(iv), and Measurement 7.5 is the controlled experiment that says so.

(L7) **Draft scope.** In this revision Sections 2–4 are complete with proofs; Sections 5–9 carry final statements with proof sketches and placeholders as marked. Table 1 is the binding record of what is claimed at what strength.

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